

6

Fourier Integrals and Fourier Transforms

6.1 Introduction

We had studied Fourier series in third semester, and we know that the series works well in treating problems involving periodic functions. If a function is not periodic, it cannot be represented by a Fourier series. Nevertheless, in such a case the function can be represented by an infinite integral. Such integrals are called Fourier integrals. Now, we shall focus our attention on: how to obtain Fourier integral from Fourier series. We shall also study Fourier cosine and sine transforms. Furthermore, the Fourier transform will also be dealt.

6.2 Fourier Integral from Fourier Series

Let $f_L(x)$ be a periodic function of period $2L$. It can be represented by Fourier series:

$$\begin{aligned} f_L(x) &= a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \\ &= \frac{1}{2L} \int_{-L}^L f_L(v) dv + \sum_{n=1}^{\infty} \left[\cos \frac{n\pi x}{L} \frac{1}{L} \int_{-L}^L f_L(v) \cos \frac{n\pi v}{L} dv + \sin \frac{n\pi x}{L} \frac{1}{L} \int_{-L}^L f_L(v) \sin \frac{n\pi v}{L} dv \right] \\ &= \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{L} \sum_{n=1}^{\infty} \left[\cos \frac{n\pi x}{L} \int_{-L}^L f_L(v) \cos \frac{n\pi v}{L} dv + \sin \frac{n\pi x}{L} \int_{-L}^L f_L(v) \sin \frac{n\pi v}{L} dv \right] \end{aligned}$$

Let $w_n = \frac{n\pi}{L}$ so that $w_{n+1} = \frac{(n+1)\pi}{L}$; and set

$$\Delta w = w_{n+1} - w_n = (n+1)\frac{\pi}{L} - \frac{n\pi}{L} = \frac{\pi}{L}. \text{ Then}$$

$$\frac{1}{L} = \frac{\Delta w}{\pi}; \text{ and } \Delta w \rightarrow 0 \text{ as } L \rightarrow \infty.$$

$$\therefore f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{\pi} \sum_{n=1}^{\infty} [\cos w_n x \int_{-L}^L f_L(v) \cos w_n v dv + \sin w_n x \int_{-L}^L f_L(v) \sin w_n v dv] \Delta w$$

Suppose that $f(x)$ is a non-periodic function which is absolutely integrable on x -axis. Let $L \rightarrow \infty$; and assume that $\lim_{L \rightarrow \infty} f_L(x) = f(x)$.

Letting $L \rightarrow \infty$, and replacing Δw by dw and Σ by the integral, $\int_0^{\infty}$,

we have

$$f(x) = \frac{1}{\pi} \int_0^{\infty} [\cos wx \int_{-\infty}^{\infty} f(v) \cos wv dv + \sin wx \int_{-\infty}^{\infty} f(v) \sin wv dv] dw$$

$$f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw, \quad (1)$$

$$\text{where } A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv$$

The integral on the right of (1) is called a **Fourier integral**.

Now, we shall state a theorem without proof.

Theorem. *If $f(x)$ is piecewise continuous in every finite interval and has a right-hand derivative and a left-hand derivative at every point and if $f(x)$ is absolutely integrable, then $f(x)$ can be*

represented by the integral $\int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw$.

At a point where $f(x)$ is discontinuous the value of the Fourier integral is equal to the average of the left and right hand limits of $f(x)$ at that point.

6.3 Fourier Cosine and Sine Integrals

1. If $f(x)$ is an even function, then

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv, \text{ and}$$

$$B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv \, dv = \frac{1}{\pi} \times 0 = 0$$

The integral (1) then reduces to

$$f(x) = \int_0^{\infty} A(w) \cos wx \, dw, \quad (2)$$

$$\text{where } A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv$$

The integral (2) is called a **Fourier cosine integral**.

2. If $f(x)$ is an odd function, then

$$A(w) = 0, \text{ and } B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv.$$

The integral (1) reduces to

$$f(x) = \int_0^{\infty} B(w) \sin wx \, dw \quad (3)$$

$$\text{where } B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv$$

The integral (3) is known as **Fourier sine integral**.

Examples

1. Choosing suitable function, show that

$$\int_0^{\infty} \frac{(\cos xw + w \sin xw)}{1+w^2} \, dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

The given integral is the Fourier integral. The Fourier integral is given by

$$f(x) = \int_0^{\infty} [A(w) \cos xw + B(w) \sin xw] dw \quad (1)$$

$$\text{Choose, } f(x) = \begin{cases} 0 & \text{if } x < 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases} \quad (2)$$

Now, we shall find $A(w)$ and $B(w)$.

$$\begin{aligned} A(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv = \frac{1}{\pi} \left[\int_{-\infty}^0 f(v) \cos wv dv + \int_0^{\infty} f(v) \cos wv dv \right] \\ &= \frac{1}{\pi} \int_0^{\infty} \pi e^{-v} \cos wv dv = \int_0^{\infty} e^{-v} \cos wv dv \\ &= e^{-v} \left\{ \frac{-\cos wv + w \sin wv}{1+w^2} \right\} \Big|_0^{\infty} \\ &= \frac{1}{1+w^2} \end{aligned}$$

$$\begin{aligned} B(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv = \frac{1}{\pi} \left[\int_{-\infty}^0 f(v) \sin wv dv + \int_0^{\infty} f(v) \sin wv dv \right] \\ &= \frac{1}{\pi} \int_0^{\infty} \pi e^{-v} \sin wv dv = \int_0^{\infty} e^{-v} \sin wv dv \\ &= e^{-v} \left\{ \frac{-\sin wv - w \cos wv}{1+w^2} \right\} \Big|_0^{\infty} \\ &= \frac{w}{1+w^2} \end{aligned}$$

Now, (1) reduces to

$$\begin{aligned} &\int_0^{\infty} \left[\frac{1}{1+w^2} \cos xw + \frac{w}{1+w^2} \sin xw \right] dw = f(x) \\ \text{or, } &\int_0^{\infty} \frac{(\cos xw + w \sin xw)}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases} \quad (3) \end{aligned}$$

The function given by (2) is not defined at $x = 0$; so it is discontinuous at $x = 0$.

At $x = 0$, the value of the Fourier integral is equal to the average of the left and right hand limits of $f(x)$ at $x=0$. Left hand limit at $x = 0$

$$\text{is } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (0) = 0$$

Right hand limit at $x=0$ is $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \pi e^{-x} = \pi$

Average of the limits $= \frac{0+\pi}{2} = \frac{\pi}{2}$

$$\therefore \int_0^{\infty} \frac{\cos xw + w \sin xw}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

2. Show that $\int_0^{\infty} \left(\frac{1-\cos \pi w}{w} \right) \sin xw dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$

The given integral is the Fourier sine integral. The Fourier sine integral is given by

$$f(x) = \int_0^{\infty} B(w) \sin xw dw \quad (1)$$

$$\text{Choose, } f(x) = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$\begin{aligned} \text{Here, } B(w) &= \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv dv = \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin wv dv = \int_0^{\pi} \sin wv dv \\ &= -\frac{1}{w} \cos wv \Big|_0^{\pi} = -\frac{1}{w} (\cos w\pi - 1) = \frac{1-\cos \pi w}{w} \end{aligned}$$

Inserting it into (1), we obtain

$$\int_0^{\infty} \left(\frac{1-\cos \pi w}{w} \right) \sin xw dw = f(x) = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

3. Represent the following function by Fourier Sine integral

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

The Fourier sine integral is

$$f(x) = \int_0^{\infty} B(w) \sin xw \, dw \quad (1)$$

$$\begin{aligned} \text{Here, } B(w) &= \frac{2}{\pi} \int_0^a f(v) \sin wv \, dv = \frac{2}{\pi} \int_0^a \sin wv \, dv \\ &= \frac{-2}{\pi} \cdot \frac{1}{w} \cos wv \Big|_0^a = -\frac{2}{w\pi} (\cos wa - 1) = \frac{2(1 - \cos wa)}{w\pi} \end{aligned}$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \left(\frac{1 - \cos wa}{w} \right) \sin xw \, dw.$$

4. Obtain the Fourier Cosine integral representation of the function $f(x) = e^{-x} + e^{-2x}$

The Fourier cosine integral is

$$f(x) = \int_0^{\infty} A(w) \cos xw \, dw \quad (1)$$

$$\text{Here, } A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv = \frac{2}{\pi} \int_0^{\infty} (e^{-v} + e^{-2v}) \cos wv \, dv$$

$$\text{Now, } \int_0^{\infty} e^{-v} \cos wv \, dv = \frac{e^{-v}(-\cos wv + w \sin wv)}{1 + w^2} \Big|_0^{\infty} = \frac{1}{1 + w^2}$$

$$\int_0^{\infty} e^{-2v} \cos wv \, dv = e^{-2v} \frac{(-2 \cos wv + w \sin wv)}{4 + w^2} \Big|_0^{\infty} = \frac{2}{4 + w^2}$$

$$\therefore A(w) = \frac{2}{\pi} \left(\frac{1}{1 + w^2} + \frac{2}{4 + w^2} \right)$$

Inserting the value of $A(w)$ in (1), we obtain

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \left(\frac{1}{1 + w^2} + \frac{2}{4 + w^2} \right) \cos xw \, dw$$

5. Find the Fourier cosine integral representation of the function

$$f(x) = \begin{cases} a^2 - x^2 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

The Fourier cosine integral is

$$f(x) = \int_0^{\infty} A(w) \cos xw \, dw$$

$$\begin{aligned}
 \text{Now, } A(w) &= \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv \\
 &= \frac{2}{\pi} \int_0^{\infty} (a^2 - v^2) \cos wv \, dv \\
 &= \frac{2}{\pi} \left[\int_0^a a^2 \cos wv \, dv - \int_0^a v^2 \cos wv \, dv \right] \\
 &= \frac{2}{\pi} \left\{ \left[a^2 \frac{\sin wv}{w} \right]_0^a - \left[v^2 \left(\frac{\sin wv}{w} \right) - 2v \left(-\frac{\cos wv}{w^2} \right) + 2 \left(-\frac{\sin wv}{w^3} \right) \right]_0^a \right\} \\
 &= \frac{2}{\pi} \left[\frac{a^2 \sin wa}{w} - \frac{a^2 \sin wa}{w} - \frac{2a \cos wa}{w^2} + \frac{2 \sin wa}{w^3} \right] \\
 &= \frac{2}{\pi} \left[\frac{2 \sin wa}{w^3} - \frac{2a \cos wa}{w^2} \right]
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 f(x) &= \frac{2}{\pi} \times 2 \int_0^{\infty} \left[\frac{\sin wa}{w^3} - \frac{a \cos wa}{w^2} \right] \cos wx \, dw \\
 &= \frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin wa}{w^3} - \frac{a \cos wa}{w^2} \right] \cos wx \, dw
 \end{aligned}$$

6. Prove $\int_0^{\infty} \frac{w^3 \sin xw}{w^4 + 4} \, dw = \frac{\pi}{2} e^{-x} \cos x$ if $x > 0$.

This is the Fourier Sine integral. It is given by

$$f(x) = \int_0^{\infty} B(w) \sin xw \, dw \quad (1)$$

$$\text{Here, } B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv$$

$$\text{Let } f(x) = \frac{\pi}{2} e^{-x} \cos x \quad (x > 0)$$

$$\begin{aligned}
 \therefore B(w) &= \frac{2}{\pi} \int_0^{\infty} \frac{\pi}{2} e^{-v} \cos v \sin wv \, dv \\
 &= \int_0^{\infty} e^{-v} \cos v \sin wv \, dv
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{\infty} e^{-v} [\sin(1+w)v + \sin(w-1)v] dv \\
 &= \frac{1}{2} \left[\int_0^{\infty} e^{-v} \sin(1+w)v dv + \int_0^{\infty} e^{-v} \sin(w-1)v dv \right] \\
 &= \frac{1}{2} \left[\frac{e^{-v}}{1+(1+w)^2} \{-\sin(1+w)v - (1+w) \cos(1+w)v\} \right]_0^{\infty} \\
 &\quad + \frac{e^{-v}}{1+(w-1)^2} \{-\sin(w-1)v - (w-1) \cos(w-1)v\} \Big|_0^{\infty} \\
 &= \frac{1}{2} \left[\frac{1+w}{1+(1+w)^2} + \frac{w-1}{1+(w-1)^2} \right] \\
 &= \frac{1}{2} \frac{(1+w)[1+(w-1)^2] + (w-1)[1+(1+w)^2]}{[1+(1+w)^2][1+(w-1)^2]} \\
 &= \frac{1}{2} \frac{2w^3}{w^4+4}
 \end{aligned}$$

$$\therefore B(w) = \frac{w^3}{w^4+4}$$

Now replacing the value of $B(w)$ in (1), we get

$$f(x) = \int_0^{\infty} \left(\frac{w^3}{w^4+4} \right) \sin wx dw$$

$$\therefore \int_0^{\infty} \left(\frac{w^3}{w^4+4} \right) \sin wx dw = \frac{\pi}{2} e^{-x} \cos x \text{ for } x > 0.$$

7. Find the Fourier cosine integral representation of $f(x) = e^{-kx}$ $x > 0$, k be a positive number.

The Fourier Cosine integral is

$$f(x) = \int_0^{\infty} A(w) \cos wx dw \quad (1)$$

$$\text{where } A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv dv$$

$$\begin{aligned}
 &= \frac{2}{\pi} \int_0^{\infty} e^{-kv} \cos wv \, dv \\
 &= \frac{2}{\pi} \left[\frac{e^{-kv}}{(k^2 + w^2)} (-k \cos wv + w \sin wv) \right]_0^{\infty} \\
 &= -\frac{2}{\pi} \frac{1}{(k^2 + w^2)} (-k)
 \end{aligned}$$

$$\therefore A(w) = \frac{2k}{\pi(k^2 + w^2)}$$

Now from eqn (1), we get

$$f(x) = \int_0^{\infty} \frac{2k}{\pi(k^2 + w^2)} \cos wx \, dw = \frac{2k}{\pi} \int_0^{\infty} \frac{\cos wx}{k^2 + w^2} \, dw$$

Some Important Results

1. $\int_0^{\infty} \frac{\sin ax}{x} \, dx = \frac{\pi}{2}$
2. $\int_0^{\infty} e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$
3. $\int_0^{\infty} e^{-ax^2} \, dx = \frac{1}{2} \frac{\sqrt{\pi}}{a}$

8. Show that

$$\int_0^{\infty} \frac{\sin \pi w \sin xw}{1 - w^2} \, dw = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 < x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

The given integral is sine integral

$$f(x) = \int_0^{\infty} B(w) \sin wx \, dw \quad (1)$$

$$\text{Choose, } f(x) = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 < x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$\text{Here, } B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv$$

$$\begin{aligned}
 \text{or, } B(w) &= \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin v \sin wv \, dv \\
 &= \int_0^{\pi} \sin v \sin wv \, dv \\
 &= \frac{1}{2} \int_0^{\pi} [\cos(wv - v) - \cos(wv + v)] \, dv \\
 &= \frac{1}{2} \left[\frac{\sin(w-1)v}{w-1} - \frac{\sin(w+1)v}{w+1} \right]_0^{\pi} \\
 &= \frac{1}{2} \left[\frac{\sin(w-1)\pi}{w-1} - \frac{\sin(w+1)\pi}{w+1} \right] \\
 &= \frac{1}{2} \left[\frac{-\sin\pi w}{w-1} + \frac{\sin\pi w}{w+1} \right] \\
 &= \frac{\sin\pi w}{2} \left[\frac{1+w+1-w}{1-w^2} \right] = \frac{\sin\pi w}{1-w^2}
 \end{aligned}$$

$$\therefore f(x) = \int_0^{\infty} \frac{\sin\pi w}{1-w^2} \sin xw \, dw$$

$$\text{or, } \int_0^{\infty} \frac{\sin\pi w}{1-w^2} \sin xw \, dw = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 < x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

Exercise 6.1

1. Choosing suitable functions, show that

$$\text{a. } \int_0^{\infty} \frac{\sin w \cos xw}{w} \, dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 \leq x < 1 \\ \frac{\pi}{4} & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$\text{b. } \int_0^{\infty} \frac{1 - \cos\pi w}{w} \sin xw \, dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$\text{c. } \int_0^{\infty} \frac{\cos \frac{\pi w}{2} \cos xw}{1-w^2} \, dw = \begin{cases} \frac{\pi}{2} \cos x \\ 0 \end{cases}$$

$$\text{d. } \int_0^{\infty} \frac{\cos xw}{1+w^2} \, dw = \frac{\pi}{2} e^{-x} \text{ if } x > 0$$

$$\text{e. } \int_0^{\infty} \frac{\sin w}{w} \cos xw \, dw = \begin{cases} \frac{\pi}{2} & \text{if } x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$\text{f. } \int_0^{\infty} \frac{w \sin wx}{a^2 + w^2} \, dw = \frac{\pi}{2} e^{-ax} \text{ for } x > 0$$

$$\text{g. } \int_0^{\infty} \frac{\cos xw}{k^2 + w^2} \, dw = \frac{\pi e^{-kx}}{2k} \text{ when } k > 0$$

$$\text{h. } \int_0^{\infty} \frac{\sin\pi w \cos xw}{w} \, dw = \begin{cases} \frac{\pi}{2} & \text{if } x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

2. Find the Fourier cosine integral of the functions.

$$\text{a. } f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$\text{b. } f(x) = e^{-x} + e^{-2x} \text{ for } x > 0$$

$$\text{c. } f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$\text{d. } f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$\text{e. } f(x) = \frac{1}{1+x^2} \text{ if } x > 0$$

3. Find the Fourier sine integral of the functions

$$\text{a. } f(x) = \begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$\text{b. } f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$\text{c. } f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$\text{d. } f(x) = \begin{cases} \pi - x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$c. \int_0^{\infty} \frac{\cos \frac{\pi w}{2} \cos xw}{1-w^2} dw = \begin{cases} \frac{\pi}{2} \cos x & \text{if } |x| < \frac{\pi}{2} \\ 0 & \text{if } |x| > \frac{\pi}{2} \end{cases}$$

$$d. \int_0^{\infty} \frac{\cos xw}{1+w^2} dw = \frac{\pi}{2} e^{-x} \text{ if } x > 0$$

$$e. \int_0^{\infty} \frac{\sin w}{w} \cos wx dw = \begin{cases} \frac{\pi}{2} & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

$$f. \int_0^{\infty} \frac{w \sin wx}{a^2 + w^2} dw = \frac{\pi}{2} e^{-ax} \text{ for } x > 0, a > 0$$

$$g. \int_0^{\infty} \frac{\cos xw}{k^2 + w^2} dw = \frac{\pi e^{-kx}}{2k} \text{ when } k > 0, x > 0$$

$$h. \int_0^{\infty} \frac{\sin \pi w \cos xw}{w} dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 \leq x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

2. Find the Fourier cosine integral representation of the following functions.

$$a. f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$b. f(x) = e^{-x} + e^{-2x} \text{ for } x > 0$$

$$c. f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$d. f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$e. f(x) = \frac{1}{1+x^2} \text{ if } x > 0$$

3. Find the Fourier sine integral representation of the followings.

$$a. f(x) = \begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$b. f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$c. f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$d. f(x) = \begin{cases} \pi - x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$e. \quad f(x) = \begin{cases} e^x & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$f. \quad f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$g. \quad f(x) = \begin{cases} 0 & \text{if } 0 < x < 1 \\ k & \text{if } 1 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$$

$$h. \quad f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$i. \quad f(x) = x^2, 0 \leq x \leq 1$$

4. Find Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

Answers

$$2. \quad a. \quad f(x) = \int_0^{\infty} \frac{2}{\pi} \left[\left(\frac{w^2 + 2}{w^3} \sin w + \frac{2}{w^2} \cos w \right) \right] \cos xw \, dw$$

$$b. \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\frac{1}{1 + w^2} + \frac{2}{2 + w^2} \right] \cos xw \, dw$$

$$c. \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin w \cos xw}{w} \, dw$$

$$d. \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\frac{a \sin aw}{w} + \frac{\cos aw - 1}{w^2} \right] \cos xw \, dw$$

$$e. \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos wx}{1 + x^2} \, dx \text{ for } w > 0$$

$$3. \quad a. \quad \frac{2}{\pi} \int_0^{\infty} \left(1 - \frac{\cos aw}{w} \right) \sin xw \, dw$$

$$b. \quad \frac{2}{\pi} \int_0^{\infty} \left[\frac{\sin aw}{w^2} - \frac{a}{w} \cos aw \right] \sin xw \, dw$$

$$c. \quad \frac{2}{\pi} \int_0^{\infty} \left(\frac{\sin \pi w}{1 - w^2} \right) \sin xw \, dw$$

d. $\frac{2}{\pi} \int_0^{\infty} \left[\frac{1}{w} - \frac{\sin w}{w^3} \right] \sin xw \, dw$

e. $\frac{2}{\pi} \int_0^{\infty} \left[\frac{w - w e^{\cos w} - \cos w}{(1+w^2)} \right] \sin xw \, dw$

f. $\frac{2}{\pi} \int_0^{\infty} \left[\frac{w e^{\cos w} - \sin w + w \cos w}{(1+w^2)e} \right] \sin xw \, dw$

g. $\frac{2k}{\pi} \int_0^{\infty} \left(\frac{\cos w - \cos 2w}{w} \right) \sin xw \, dw$

h. Problem for the reader

i. $\frac{2}{\pi} \int_0^{\infty} \left[\left(-\frac{1}{w} + \frac{2}{w^3} \right) \cos w + \frac{2 \sin w}{w^2} - \frac{2}{w^3} \right] \sin xw \, dw$

4. $\frac{2}{\pi} \int_0^{\infty} \frac{\sin w \cos xw}{w} \, dw$

6.4 Fourier Cosine and Sine Transforms

By transformation we mean an operation which transforms a function into another function. If the output function appears in the form of an integral, the transform is an integral transform. This section deals with Fourier cosine and sine transforms which are nothing but integral transforms.

We are familiar with Fourier cosine and sine integrals, and the Fourier cosine integral is

$$f(x) = \int_0^{\infty} A(w) \cos xw \, dw, \text{ where } A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv$$

Setting $A(w) = \sqrt{\frac{2}{\pi}} \hat{f}_c(w)$, we have

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(v) \cos wv \, dv = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx, \text{ and}$$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_c(w) \cos xw \, dw.$$

Here, $\hat{f}_c(w)$ is called the Fourier cosine transform of $f(x)$, and $f(x)$ the inverse Fourier transform of $\hat{f}_c(w)$.

Fourier Cosine Transform

Let $f(x)$ be absolutely integrable on the positive x -axis, and piecewise continuous on every finite interval.

The Fourier cosine transform of $f(x)$, denoted by $\hat{f}_c(w)$, is defined by

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx \quad - (1)$$

and, the inverse Fourier cosine transform is defined by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_c(w) \cos xw \, dw \quad - (2)$$

Observation: In (1) we observe that the function $f(x)$ is transformed into a new function $\hat{f}_c(w)$, and in (2) $\hat{f}_c(w)$ is transformed into the original function.

Here, we call $\hat{f}_c(w)$ the Fourier cosine transform of $f(x)$; and $f(x)$ the inverse Fourier cosine transform. The other notation commonly used is $\mathfrak{F}_c(f)$. The inverse of $\mathfrak{F}_c$ is $\mathfrak{F}_c^{-1}$.

The Fourier sine integral is

$$f(x) = \int_0^{\infty} B(w) \sin xw \, dw, \text{ where } B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv$$

Setting $B(w) = \sqrt{\frac{2}{\pi}} \hat{f}_s(w)$, we obtain

$$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(v) \sin wv \, dv = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx.$$

Here, $\hat{f}_s(w)$ is called the Fourier sine transform of $f(x)$, and $f(x)$ the inverse Fourier sine transform.

Fourier Sine Transform

Let $f(x)$ be absolutely integrable on the positive x -axis, and piecewise continuous on every finite interval.

The Fourier sine transform, denoted by $\hat{f}_s$, is defined by

$$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx, \quad (3)$$

and the inverse Fourier sine transform is defined by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_s(w) \sin xw \, dw \quad (4)$$

$\hat{f}_s$ is denoted by $\mathfrak{F}_s(f)$ and the inverse of $\mathfrak{F}_s$ is $\mathfrak{F}_s^{-1}$.

6.5 Properties of Fourier Sine and Cosine Transform

1. **Linearity.** If $f(x)$ is absolutely integrable on the x -axis, and piece wise continuous on every finite interval, then

$$\mathfrak{F}_c[af(x) + bg(x)] = a \mathfrak{F}_c[f(x)] + b \mathfrak{F}_c[g(x)] \quad (i)$$

$$\mathfrak{F}_s[af(x) + bg(x)] = a \mathfrak{F}_s[f(x)] + b \mathfrak{F}_s[g(x)] \quad (ii)$$

where a and b are constants.

$$\begin{aligned} \text{Proof. } \mathfrak{F}_c[af(x) + bg(x)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} [af(x) + bg(x)] \cos wx \, dx \\ &= a \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx \right] + b \left[\sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \cos wx \, dx \right] \\ &= a \mathfrak{F}_c[f(x)] + b \mathfrak{F}_c[g(x)]. \end{aligned}$$

Similarly, we can prove (ii), and it is left for exercise.

2. Cosine and Sine Transforms of Derivatives

Let $f(x)$ be continuous and absolutely integrable on the x -axis, and $f'(x)$ be piecewise continuous on each finite interval, and let $f(x) \rightarrow 0$ as $x \rightarrow \infty$; then

$$\mathfrak{I}_c[f'(x)] = w\mathfrak{I}_s[f(x)] - \sqrt{\frac{2}{\pi}} f(0) \quad (i)$$

$$\mathfrak{I}_s[f'(x)] = -w\mathfrak{I}_c[f(x)] \quad (ii)$$

$$\mathfrak{I}_c[f''(x)] = -w^2\mathfrak{I}_c[f(x)] - \sqrt{\frac{2}{\pi}} f'(0) \quad (iii)$$

$$\mathfrak{I}_s[f''(x)] = -w^2\mathfrak{I}_s[f(x)] + \sqrt{\frac{2}{\pi}} w f(0) \quad (iv)$$

Proof. (i) By definition, $\mathfrak{I}_c[f'(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f'(x) \cos wx \, dx$

$$= \sqrt{\frac{2}{\pi}} \left[\cos wx \cdot f(x) \Big|_0^{\infty} + w \int_0^{\infty} \sin wx \cdot f(x) dx \right]$$

Since $f(x) \rightarrow 0$ as $x \rightarrow \infty$, so the first term reduces to $-\sqrt{\frac{2}{\pi}} f(0)$

$$\therefore \mathfrak{I}_c[f'(x)] = w\mathfrak{I}_s[f(x)] - \sqrt{\frac{2}{\pi}} f(0)$$

$$(ii) \mathfrak{I}_s[f'(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f'(x) \sin wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\sin wx \cdot f(x) \Big|_0^{\infty} - w \int_0^{\infty} \cos wx \cdot f(x) dx \right]$$

(on integrating by parts)

Since $f(x) \rightarrow 0$ as $x \rightarrow \infty$, so the first term vanishes.

$$\therefore \mathfrak{I}_s[f'(x)] = -w\mathfrak{I}_c[f(x)].$$

$$(iii) \mathfrak{I}_c[f''(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f''(x) \cos wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\cos wx \cdot f'(x) \Big|_0^{\infty} + w \int_0^{\infty} \sin wx \cdot f'(x) dx \right]$$

The first term reduces to $-\sqrt{\frac{2}{\pi}} f'(0)$.

$$\therefore \mathfrak{I}_c[f''(x)] = -\sqrt{\frac{2}{\pi}} f'(0) + w\mathfrak{I}_s[f'(x)]$$

QUESTION 1: Find the area of the region bounded by the curve $y = \sqrt{x}$ and the line $y = x$.

$$y = \sqrt{x} \quad \text{and} \quad y = x$$

Solution: The curves intersect at $(0,0)$ and $(1,1)$.

$$\text{Area} = \int_0^1 (\sqrt{x} - x) dx$$

Integrating with respect to x ,

$$= \left[\frac{2}{3}x^{3/2} - \frac{1}{2}x^2 \right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

ANSWER: $\frac{1}{6}$

QUESTION 2: Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

$$y = x^2 \quad \text{and} \quad y = 2x$$

$$\text{Area} = \int_0^2 (2x - x^2) dx$$

$$= \left[x^2 - \frac{1}{3}x^3 \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

ANSWER: $\frac{4}{3}$

$$\text{Area} = \int_0^1 (1 - x^2) dx$$

$$= \left[x - \frac{1}{3}x^3 \right]_0^1 = 1 - \frac{1}{3} = \frac{2}{3}$$

$$= \frac{2}{3}$$

$$\left[\frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

2. Find the Fourier sine and cosine transform of $f(t) = e^{-at}$.

The Fourier sine transform of $f(t)$ is

$$\mathfrak{F}_s[f(t)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin wt \, dt$$

$$\text{or, } \mathfrak{F}_s(e^{-at}) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \sin wt \, dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-at}}{a^2 + w^2} (-a \sin wt - w \cos wt) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[0 - \frac{1}{a^2 + w^2} (-w) \right]$$

$$\therefore \mathfrak{F}_s(e^{-at}) = \sqrt{\frac{2}{\pi}} \left(\frac{w}{a^2 + w^2} \right)$$

Also, the Fourier cosine transform of $f(t)$ is

$$\mathfrak{F}_c[f(t)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos wt \, dt$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \cos wt \, dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-at}}{a^2 + w^2} (-a \cos wt + w \sin wt) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[0 + \frac{a}{a^2 + w^2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + w^2} \right)$$

$$\therefore \mathfrak{F}_c[f(t)] = \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + w^2} \right)$$

3. Find the Fourier cosine transform of $f(t) = e^{-t^2}$.

By definition, we have

$$\mathfrak{F}_c[f(t)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos wt \, dt$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t^2} \cos wt \, dt$$

$$\text{Let } F = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t^2} \cos wt \, dt \quad (1)$$

Now differentiating both sides with respect to w , we get

$$\frac{dF}{dw} = -\sqrt{\frac{2}{\pi}} \int_0^{\infty} t e^{-t^2} \sin wt \, dt$$

$$= -\frac{1}{2} \sqrt{\frac{2}{\pi}} \int_0^{\infty} 2t e^{-t^2} \sin wt \, dt$$

$$= -\frac{1}{2} \sqrt{\frac{2}{\pi}} \int_0^{\infty} (e^{-t^2})' \sin wt \, dt$$

$$= -\frac{1}{2} \sqrt{\frac{2}{\pi}} \left\{ [\sin wt e^{-t^2}]_0^{\infty} - w \int_0^{\infty} \cos wt e^{-t^2} dt \right\}$$

$$= -\frac{w}{2} \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t^2} \cos wt \, dt \quad [\because \text{The first term vanishes}]$$

$$= -\frac{w}{2} F$$

$$\text{or, } \frac{dF}{dw} = -\frac{wF}{2}$$

$$\text{or, } \frac{dF}{F} = -\frac{1}{2} w \, dw$$

On integration

$$\log F = -\frac{w^2}{4} + \log C$$

$$\text{or, } \log\left(\frac{F}{C}\right) = -\frac{w^2}{4}$$

$$\text{or, } F = C e^{-\frac{w^2}{4}} \quad (2)$$

Initially when $w = 0$

$$F = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t^2} dt = \sqrt{\frac{2}{\pi}} \frac{\sqrt{\pi}}{2} = \frac{1}{\sqrt{2}}$$

Putting the value of F in (2), $C = \frac{1}{\sqrt{2}}$

$$\therefore F = \mathfrak{F}_s[f(t)] = \frac{1}{\sqrt{2}} e^{-\frac{w^2}{4}}$$

4. Find the Fourier sine transform of e^{-x} and then prove that

$$\int_0^{\infty} \frac{w \sin wx}{1 + w^2} dw = \frac{\pi}{2} e^{-x}.$$

By definition, the Fourier sine transform is

$$\begin{aligned} \mathfrak{F}_s[f(x)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \sin wx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-x}}{1 + w^2} (-\sin wx - w \cos wx) \right]_0^{\infty} \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{w}{1 + w^2} \right) \end{aligned}$$

$$\therefore \mathfrak{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \left(\frac{w}{1 + w^2} \right)$$

Again by inverse Fourier sine transform,

$$\begin{aligned} f(x) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \mathfrak{F}_s[f(x)] \sin wx \, dw \\ \text{or, } e^{-x} &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \left(\frac{w}{1 + w^2} \right) \sin wx \, dw \\ &= \frac{2}{\pi} \int_0^{\infty} \left(\frac{w \sin wx}{1 + w^2} \right) dw \\ \therefore \int_0^{\infty} \left(\frac{w \sin wx}{1 + w^2} \right) dw &= \frac{\pi}{2} e^{-x} \end{aligned}$$

5. Find Fourier Cosine transform of $f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$

The Fourier cosine transform of $f(x)$ is

$$\begin{aligned}\mathfrak{F}_c[f(x)] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx = \sqrt{\frac{2}{\pi}} \int_0^{\infty} x \cos wx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[\frac{x \sin wx}{w} + \frac{1}{w^2} \cos wx \right]_0^{\infty} \\ &= \sqrt{\frac{2}{\pi}} \left[\frac{a \sin wa}{w} + \frac{\cos wa}{w^2} - \frac{1}{w^2} \right]\end{aligned}$$

6. Find $\mathfrak{F}_c(e^{-\frac{1}{2}x^2})$

$$\text{Let } I = \mathfrak{F}_c(e^{-\frac{1}{2}x^2}) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{1}{2}x^2} \cos wx \, dx \quad (1)$$

$$\begin{aligned}\frac{dI}{dw} &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} (-x) e^{-\frac{1}{2}x^2} \sin wx \, dx = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \left(e^{-\frac{1}{2}x^2} \right)' \sin wx \, dx \\ &= \mathfrak{F}_s \left(e^{-\frac{1}{2}x^2} \right)' = -w \mathfrak{F}_c(e^{-\frac{1}{2}x^2}) = -wI\end{aligned}$$

$$\text{Thus, } \frac{dI}{dw} = -wI \text{ or, } \frac{dI}{I} = -w dw$$

$$\text{On integration, } \ln I = -\frac{w^2}{2} + \ln c \text{ or } \ln(I/c) = -\frac{w^2}{2}$$

$$\text{or } \frac{I}{c} = e^{-\frac{w^2}{2}}$$

$$I = c e^{-\frac{w^2}{2}} \quad (2)$$

$$\text{Initially when } w = 0, \text{ then from (1), } I = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{1}{2}x^2} \, dx$$

$$\text{or, } I = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{2} \sqrt{2\pi} = 1$$

$$\text{From (2), } I = c e^0 \quad \text{or} \quad 1 = c$$

$$z.0 = e^{-\frac{x^2}{2}}$$

7. Find $\mathcal{L}_s(e^{-\frac{x^2}{2}})$

By definition, $\mathcal{L}_s(e^{-\frac{x^2}{2}})$

$$= -\sqrt{\frac{\pi}{2}} \int_0^\infty x e^{-\frac{x^2}{2}} \sin wx \, dx$$

$$= -\sqrt{\frac{\pi}{2}} \int_0^\infty x e^{-\frac{x^2}{2}} \sin wx \, dx$$

$$= -\mathcal{L}_s(x e^{-\frac{x^2}{2}})$$

$$= -(-w) \mathcal{L}_s(e^{-\frac{x^2}{2}})$$

$$= w e^{-\frac{w^2}{2}}$$

(By example)

Exercise 6.

1. Find the Fourier Cosine and Sine

a. $f(x) = 2e^{-x} + 3e^{-2x}$

b. $f(x) = 2e^{-x} + 5e^{-2x}$

c. $f(x) = \begin{cases} 1 & 0 < x < a \\ 0 & x > a \end{cases}$

d. $f(x) = \frac{e^{-ax}}{a}$

e. $f(x) = ae^{-ax}$

2. Find Fourier Cosine transform of

a. $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2-x & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$

b. $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 1-x & \text{for } \frac{1}{2} < x < 1 \\ 0 & \text{for } x > 1 \end{cases}$

c. $f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ -1 & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$

3. Find Fourier Cosine and Sine

$$f(x) = \begin{cases} x^2 & \text{for } 0 < x < 1 \\ 0 & \text{for } x > 1 \end{cases}$$

4. Find Fourier Cosine transform

$$f(x) = \begin{cases} \cos x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

5. Find Fourier Sine transform

$$f(x) = \left(\frac{x^2}{x^2 + 4} \right) \quad x > 0$$

6. Find the Fourier Cosine transform

(a) $f(x) = x^n$ (b) $f(x) = e^{-x^2}$

Exercise 6.2

1. Find the Fourier Cosine and Sine transforms of the followings.

a. $f(x) = 2e^{-3x} + 3e^{-2x}$

b. $f(x) = 2e^{-5x} + 5e^{-2x}$

c. $f(x) = \begin{cases} 1 & 0 < x < a \\ 0 & x > a \end{cases}$

d. $f(x) = \frac{e^{-ax}}{x}$

e. $f(x) = xe^{-ax}$

2. Find Fourier Cosine transform of

a. $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2-x & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$

b. $f(x) = \begin{cases} x & \text{for } 0 < x < \frac{1}{2} \\ 1-x & \text{for } \frac{1}{2} < x < 1 \\ 0 & \text{for } x > 1 \end{cases}$

c. $f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ -1 & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$

3. Find Fourier Cosine and Sine transform of

$$f(x) = \begin{cases} x^2 & \text{for } 0 < x < 1 \\ 0 & \text{for } x > 1 \end{cases}$$

4. Find Fourier Cosine transform of

$$f(x) = \begin{cases} \cos x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

5. Find Fourier Sine transform of

$$f(x) = \left(\frac{x^3}{x^4 + 4} \right) \quad x > 0$$

6. Find the Fourier Cosine transform of the following functions

(a) $f(x) = e^{-x}$ (b) $f(x) = e^{-kx}$ ($k > 0$)

7. Find the Fourier Cosine transform of the function

$$f(x) = \begin{cases} p & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

8. Find the Fourier Cosine transform of the function

$$f(x) = \begin{cases} x & \text{if } 0 < x < b \\ 0 & \text{if } x > b \end{cases}$$

9. Find $\mathfrak{F}_c(e^{-x^2})$

10. Find Fourier sine transform of the function

$$(a) f(x) = e^{-3x} \quad (b) f(x) = e^{-kx} (k > 0)$$

11. Find the Fourier sine transform of the function

$$f(x) = \begin{cases} p & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

Answers

$$1. \quad a. \quad \sqrt{\frac{2}{\pi}} \left[\frac{6}{9+w^2} + \frac{6}{w^2+4} \right], \quad \sqrt{\frac{2}{\pi}} \left[\frac{3w}{w^2+1} + \frac{5w}{w^2+9} \right]$$

$$b. \quad \sqrt{\frac{2}{\pi}} \left[\frac{10}{w^2+25} + \frac{10}{w^2+4} \right], \quad \sqrt{\frac{2}{\pi}} \left[\frac{2w}{w^2+25} + \frac{5w}{w^2+4} \right]$$

$$c. \quad \sqrt{\frac{2}{\pi}} \frac{\sin aw}{w}, \quad \sqrt{\frac{2}{\pi}} \left(\frac{1 - \cos aw}{w} \right)$$

$$d. \quad \sqrt{\frac{2}{\pi}} \tan^{-1} \left(\frac{w}{a} \right), \quad \sqrt{\frac{2}{\pi}} \left[\log \left(\frac{1}{\sqrt{w^2+a^2}} \right) + c \right]$$

$$e. \quad \sqrt{\frac{2}{\pi}} \left[\frac{a^2 - w^2}{(a^2 + w^2)^2} \right], \quad \sqrt{\frac{2}{\pi}} \left[\frac{2aw}{(a^2 + w^2)^2} \right]$$

$$2. \quad a. \quad \sqrt{\frac{2}{\pi}} \left[\frac{2 \cos w (1 - \cos w)}{w^2} \right]$$

$$b. \quad \sqrt{\frac{2}{\pi}} \left[-\frac{\cos w}{w^2} + \frac{2 \cos \frac{w}{2}}{w^2} - \frac{1}{w^2} \right]$$

- c. $\sqrt{\frac{2}{\pi}} \left[\frac{2\sin w - \sin 2w}{w} \right]$
3. $\sqrt{\frac{2}{\pi}} \left[\frac{2w \cos w + (w^2 - 2) \sin w}{w^3} \right], \sqrt{\frac{2}{\pi}} \left[\frac{(2 - w^2) \cos w + 2w \sin w}{w^3} \right]$
4. $\sqrt{\frac{2}{\pi}} \left[\sin \frac{(1+w)a}{1+w} + \frac{\sin(1-w)a}{1-w} \right]$
5. $\sqrt{\frac{\pi}{2}} e^{-w} \cos w$
6. a. $\sqrt{\frac{2}{\pi}} \cdot \left(\frac{1}{1+w^2} \right)$
 b. $\sqrt{\frac{2}{\pi}} \left(\frac{k}{k^2 + w^2} \right)$
7. $\sqrt{\frac{2}{\pi}} p \frac{\sin w}{w}$
8. $\sqrt{\frac{2}{\pi}} \left[\frac{a \sin wa}{w} + \frac{(\cos wa - 1)}{w^2} \right]$
9. $\frac{1}{\sqrt{2}} e^{-w^2/4}$
10. a. $\sqrt{\frac{2}{\pi}} \left(\frac{w}{9+w^2} \right)$ b. $\sqrt{\frac{2}{\pi}} \left(\frac{w}{k^2 + w^2} \right)$
11. $\sqrt{\frac{2}{\pi}} p(1 - \cos wa)$

6.6 Fourier Integral in Complex Form

The Fourier integral is

$$f(x) = \int_0^{\infty} [A(w)\cos wx + B(w)\sin wx] dw \quad (1)$$

$$\left. \begin{aligned} \text{where } A(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \cos wt \, dt \\ \text{and } B(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \sin wt \, dt \end{aligned} \right\} \quad (2)$$

Now replacing the value of (2) in (1)

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) [\cos wt \cos wx + \sin wt \sin wx] \, dt \, dw \\ &= \frac{1}{\pi} \int_0^{\infty} \left[\int_{-\infty}^{\infty} f(t) \cos(wx - wt) \, dt \right] dw \\ &\quad \text{(Since cos is an even function)} \\ &= \frac{1}{2} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) \cos(wx - wt) \, dt \right] dw \quad (3) \end{aligned}$$

Since $\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) \sin(wx - wt) \, dt \right] dw$ is an odd function its value is zero. So adding zero in (3) does not alter the result. Hence

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) \cos(wx - wt) \, dt \right] dw + \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) \sin(wx - wt) \, dt \right] dw \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{i(wx - wt)} \, dt \, dw \right] \end{aligned}$$

$\therefore f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{iw(x-t)} \, dt \, dw$ which is complex Fourier integral.

6.7 Developing Fourier Transform and Its Inverse

The Fourier integral in complex form is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{i\omega(x-t)} dt d\omega$$

It can be written as

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right] e^{i\omega x} d\omega \quad (1)$$

$$\text{Let } \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (2)$$

The equation (1) then reduces to

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega \quad (3)$$

The equation (2) is said to be Fourier transform of $f(t)$ in complex form and equation (3) is its inverse.

Fourier Transform

Let $f(x)$ be absolutely integrable over $(-\infty, \infty)$; the Fourier transform of $f(x)$, denoted by $\hat{f}(\omega)$, is defined by

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx.$$

We refer to the operation that carries $f(x)$ into $\hat{f}(\omega)$ as the Fourier transformation. We shall denote it by $\mathcal{F}$. The notations $\mathcal{F}f$ and $\hat{f}(\omega)$ will be used interchangeably.

The inverse Fourier transform is defined by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega.$$

6.8 Properties of Fourier Transform

1. Linearity

Let $f(x)$ and $g(x)$ be two functions whose Fourier transforms exist, then for any constants a and b ,

$$\Im [af(x) + bg(x)] = a\Im (f) + b\Im (g)$$

Proof.

By definition,

$$\begin{aligned}\Im [af(x) + bg(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (af + bg) e^{-iwx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} af(x) e^{-iwx} dx + \int_{-\infty}^{\infty} bg(x) e^{-iwx} dx \right] \\ &= a \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \right] + b \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-iwx} dx \right] \\ &= a\Im (f) + b\Im (g)\end{aligned}$$

$$\therefore \Im (af + bg) = a\Im (f) + b\Im (g) \quad \text{proved.}$$

2. Derivative of Fourier Transform

Let $f(x)$ be continuous on x -axis such that $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and $f(x)$ is absolutely integrable in x -axis, then

$$(a) \Im [f'(x)] = iw\Im [f(x)]$$

$$\text{and (b) } \Im [f''(x)] = -w^2\Im [f(x)]$$

Proof.

(a) By definition of Fourier transform, we have

$$\begin{aligned}\Im [f'(x)] &= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} f'(x) e^{-iwx} dx \right] \\ &= \frac{1}{\sqrt{2\pi}} \left\{ [f(x) e^{-iwx}]_{-\infty}^{\infty} + iw \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \right\} \\ &= \frac{1}{\sqrt{2\pi}} [0 + iw \int_{-\infty}^{\infty} f(x) e^{-iwx} dx] \\ &= \frac{iw}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx\end{aligned}$$

$$\therefore \Im [f'(x)] = iw\Im [f(x)]$$

$$\begin{aligned}
 (b) \quad \mathcal{F}[f'(x)] &= i\omega \mathcal{F}[f(x)] \\
 &= i\omega (i\omega) \mathcal{F}[f(x)] \\
 &= i^2 \omega^2 \mathcal{F}[f(x)] \\
 &= -\omega^2 \mathcal{F}[f(x)] \\
 \therefore \mathcal{F}[f'(x)] &= -\omega^2 \mathcal{F}[f(x)]
 \end{aligned}$$

Examples

1. Find the Fourier transform of $f(x) = \begin{cases} k & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned}
 \mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_0^a k e^{-i\omega x} dx \\
 &= \frac{k}{\sqrt{2\pi}} \left(\frac{e^{-i\omega x}}{-i\omega} \right)_0^a \\
 &= \frac{k}{\sqrt{2\pi}} \left(\frac{e^{-i\omega a} - 1}{-i\omega} \right) \\
 &= \frac{k(1 - e^{-i\omega a})}{i\omega \sqrt{2\pi}}
 \end{aligned}$$

2. Find the Fourier transform of $f(x) = \begin{cases} e^x & \text{if } x < 0 \\ e^{-x} & \text{if } x > 0 \end{cases}$

The Fourier transform of $f(x)$ is given by

$$\begin{aligned}
 \mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\
 &= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^0 f(x) e^{-i\omega x} dx + \int_0^{\infty} f(x) e^{-i\omega x} dx \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^0 e^x e^{-iwx} dx + \int_0^{\infty} e^{-x} e^{-iwx} dx \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^0 e^{(1-iw)x} dx + \int_0^{\infty} e^{-(1+iw)x} dx \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left\{ \left[\frac{e^{(1-iw)x}}{1-iw} \right]_{-\infty}^0 + \left[\frac{e^{-(1+iw)x}}{-(1+iw)} \right]_0^{\infty} \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{1-iw} (1-0) - \frac{1}{1+iw} (0-1) \right] \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{1-iw} + \frac{1}{1+iw} \right] \\
 &= \frac{2}{\sqrt{2\pi}} \left(\frac{1}{1+w^2} \right) \\
 &= \sqrt{\frac{2}{\pi}} \left(\frac{1}{1+w^2} \right)
 \end{aligned}$$

3. Find the Fourier transform of the function

$$f(x) = \begin{cases} 1-x^2 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$$

The Fourier transform of $f(x)$ is given by

$$\begin{aligned}
 \mathfrak{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) e^{-iwx} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) (\cos wx - i \sin wx) dx \\
 &= \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^1 (1-x^2) \cos wx dx - i \int_{-1}^1 (1-x^2) \sin wx dx \right] \\
 &= \frac{2}{\sqrt{2\pi}} \int_0^1 (1-x^2) \cos wx dx - 0
 \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{\frac{2}{\pi}} \left[\frac{(1-x^2)\sin wx}{w} - \frac{2x\cos wx}{w^2} + \frac{2\sin wx}{w^3} \right]_0^1 \\
 &= \sqrt{\frac{2}{\pi}} \left(\frac{2\sin w}{w^3} - \frac{2\cos w}{w^2} \right) \\
 &= 2\sqrt{\frac{2}{\pi}} \left(\frac{\sin w - w\cos w}{w^3} \right)
 \end{aligned}$$

4. Find the Fourier transformation of $f(x) = e^{-\frac{x^2}{2}}$

$$\begin{aligned}
 \mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} e^{-iwx} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{x^2}{2} + iwx\right)} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left[\left(\frac{x}{\sqrt{2}} + \frac{iw}{\sqrt{2}}\right)^2 - \left(\frac{iw}{\sqrt{2}}\right)^2\right]} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(\frac{iw}{\sqrt{2}}\right)^2} e^{-\left(\frac{x+iw}{\sqrt{2}}\right)^2} dx \\
 &= \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} \int_{-\infty}^{\infty} e^{-\left(\frac{x+iw}{\sqrt{2}}\right)^2} dx
 \end{aligned}$$

Let $\frac{x+iw}{\sqrt{2}} = p$. As $x \rightarrow \infty$, $p \rightarrow \infty$, and

As $x \rightarrow -\infty$, $p \rightarrow -\infty$

Also, $dx = \sqrt{2} dp$

$$\therefore \mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} \int_{-\infty}^{\infty} e^{-p^2} \sqrt{2} dp = \frac{2}{\sqrt{\pi}} e^{-\frac{w^2}{2}} \int_0^{\infty} e^{-p^2} dp$$

$$= \frac{2}{\sqrt{\pi}} e^{-\frac{w^2}{4}} \frac{\sqrt{\pi}}{2}$$

$$= e^{-\frac{w^2}{4}}$$

Proceeding as before, we obtain $\mathfrak{F}[e^{-x^2}] = \frac{1}{\sqrt{2}} e^{-\frac{w^2}{4}}$.

5. Find the Fourier transform of $f(x) = xe^{-x^2}$.

Here $f(x) = xe^{-x^2}$ (1)

Let $p(x) = e^{-x^2}$ (2)

Differentiating (2) with respect to x

$$p'(x) = -2xe^{-x^2}$$

or, $-\frac{p'(x)}{2} = xe^{-x^2}$

Now taking Fourier transform to both sides

$$\mathfrak{F}\left(\frac{p'(x)}{-2}\right) = \mathfrak{F}(xe^{-x^2})$$

or, $-\frac{1}{2} \mathfrak{F}[p'(x)] = \mathfrak{F}[f(x)]$

or, $-\frac{iw}{2} \mathfrak{F}[p(x)] = \mathfrak{F}[f(x)]$

or, $\mathfrak{F}[f(x)] = -\frac{iw}{2} \left(\frac{1}{\sqrt{2}}\right) e^{-\frac{w^2}{4}} = -\frac{iw}{2\sqrt{2}} e^{-\frac{w^2}{4}}$

6. Find the Fourier transform of $f(x) = \begin{cases} 1 & \text{for } |x| < a \\ 0 & \text{for } |x| > a \end{cases}$

The Fourier transform of a function $f(x)$ is given by

$$\mathfrak{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^a 1 \cdot e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-iwx}}{-iw} \right]_{-a}^a$$

$$= \frac{1}{\sqrt{2\pi}} \frac{(e^{-iwa} - e^{iwa})}{-iw}$$

$$= \sqrt{\frac{2}{\pi}} \frac{\sin wa}{w}$$

7. Find the Fourier transform of

$$f(t) = \begin{cases} 2 & \text{for } -1 < t < 1 \\ 1 & \text{for } -2 < t < -1 \\ 1 & \text{for } 1 < t < 2 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform

$$\mathfrak{F}[f(t)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-iwt} dt$$

Replacing the given value

$$\mathfrak{F}[f(t)] = \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^1 2e^{-iwt} dt + \int_{-2}^{-1} 1e^{-iwt} dt + \int_1^2 1e^{-iwt} dt \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[2 \left(\frac{e^{-iwt}}{-iw} \right)_{-1}^1 + \left(\frac{e^{-iwt}}{-iw} \right)_{-2}^{-1} + \left(\frac{e^{-iwt}}{-iw} \right)_1^2 \right]$$

$$= \frac{1}{w\sqrt{2\pi}} \left[2(e^{-iw} - e^{iw}) + (e^{-iw} - e^{-2iw}) + (e^{-2iw} - e^{-iw}) \right]$$

$$= \frac{1}{w\sqrt{2\pi}} [2\sin w]$$

$$= \frac{1}{w\sqrt{2\pi}} [2\sin w]$$

$$= \frac{2}{w\sqrt{2\pi}} \sin w$$

$$= \sqrt{\frac{2}{\pi}} \frac{\sin w}{w}$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-iwx}}{-iw} \right]_a \\
 &= \frac{1}{\sqrt{2\pi}} \frac{(e^{-iwa} - e^{iwa})}{-iw} \\
 &= \sqrt{\frac{2}{\pi}} \frac{\sin wa}{w}
 \end{aligned}$$

7. Find the Fourier transform of the function

$$f(t) = \begin{cases} 2 & \text{for } -1 < t < 1 \\ 1 & \text{for } -2 < t < -1 \\ 1 & \text{for } 1 < t < 2 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of the function $f(t)$ is given by

$$\mathfrak{F}[f(t)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-iwt} dt \quad (1)$$

Replacing the given values of $f(t)$ in (1), we get

$$\begin{aligned}
 \mathfrak{F}[f(t)] &= \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^1 2e^{-iwt} dt + \int_{-2}^{-1} e^{-iwt} dt + \int_1^2 e^{-iwt} dt \right] \\
 &= \frac{1}{\sqrt{2\pi}} \left[2 \left(\frac{e^{-iwt}}{-iw} \right)_{-1}^1 + \left(\frac{e^{-iwt}}{-iw} \right)_{-2}^{-1} + \left(\frac{e^{-iwt}}{-iw} \right)_1^2 \right] \\
 &= \frac{1}{w\sqrt{2\pi}} \left[2 \left(\frac{e^{iw} - e^{-iw}}{i} \right) + \left(\frac{e^{2iw} - e^{iw}}{i} \right) - \left(\frac{e^{-iw} - e^{-2iw}}{i} \right) \right] \\
 &= \frac{1}{w\sqrt{2\pi}} \left[\left(\frac{e^{iw} - e^{-iw}}{i} \right) + \left(\frac{e^{2iw} - e^{-2iw}}{i} \right) \right] \\
 &= \frac{1}{w\sqrt{2\pi}} [2\sin w + 2\sin 2w] \\
 &= \frac{1}{w\sqrt{2\pi}} [2\sin w + 4\sin w \cos w] \\
 &= \frac{2}{w\sqrt{2\pi}} \sin w (1 + 2\cos w) \\
 &= \sqrt{\frac{2}{\pi}} \frac{\sin w}{w} (1 + 2\cos w)
 \end{aligned}$$

6.9 Convolution

The convolution $f * g$ of functions $f(x)$ and $g(x)$ is defined by

$$h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(p) g(x - p) dp = \int_{-\infty}^{\infty} f(x - p) g(p) dp$$

Convolution Theorem on Fourier transform

Let $f(x)$ and $g(x)$ be piecewise continuous, bounded and absolutely integrable on x -axis, then

$$\mathfrak{F}(f * g) = \sqrt{2\pi} [\mathfrak{F}(f) \mathfrak{F}(g)]$$

Proof.

By definition,

$$\begin{aligned} \mathfrak{F}[f * g(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (f * g)(x) e^{-iwx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x-p) e^{-iwx} dp dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x-p) e^{-iwx} dx dp \quad (\text{By changing the order of integration}) \end{aligned}$$

$$\text{Let } x - p = q \Rightarrow x = p + q \Rightarrow dx = dq$$

The limit of integration remains unchanged

$$\begin{aligned} \mathfrak{F}[f * g] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(q) e^{-i w(p+q)} dq dp \\ &= \frac{1}{\sqrt{2\pi}} \left[\left(\int_{-\infty}^{\infty} f(p) e^{-iwp} dp \right) \left(\int_{-\infty}^{\infty} g(q) e^{-iwq} dq \right) \right] \\ &= \frac{1}{\sqrt{2\pi}} \left[\left(\int_{-\infty}^{\infty} f(p) e^{-iwp} dp \right) \times \sqrt{2\pi} \times \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(q) e^{-iwq} dq \right] \\ &= \sqrt{2\pi} \mathfrak{F}(f) \cdot \mathfrak{F}(g) \end{aligned}$$

6.10 Parseval's Identity for Fourier Transform

If the Fourier transform of $f(x)$ and $g(x)$ be $\hat{f}(w)$ and $\hat{g}(w)$ respectively then

$$i. \int_{-\infty}^{\infty} \hat{f}(w) \bar{\hat{g}}(w) dw = \int_{-\infty}^{\infty} f(x) \bar{g}(x) dx$$

$$ii. \int_{-\infty}^{\infty} |\hat{f}(w)|^2 dw = \int_{-\infty}^{\infty} |f(x)|^2 dx$$

where the bar indicates the complex conjugates.

Proof:

$$\text{Here } \int_{-\infty}^{\infty} f(x) \bar{g}(x) dx = \int_{-\infty}^{\infty} f(x) \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \bar{\hat{g}}(w) e^{-iwx} dw \right] dx$$

$$= \int_{-\infty}^{\infty} \bar{\hat{g}}(w) \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \right) dw$$

$$= \int_{-\infty}^{\infty} \bar{\hat{g}}(w) \hat{f}(w) dw$$

$$\therefore \int_{-\infty}^{\infty} f(x) \bar{g}(x) dx = \int_{-\infty}^{\infty} \hat{f}(w) \bar{\hat{g}}(w) dw$$

$$b. \text{ Again } \int_{-\infty}^{\infty} |\hat{f}(w)|^2 dw = \int_{-\infty}^{\infty} \hat{f}(w) \bar{\hat{f}}(w) dw = \int_{-\infty}^{\infty} f(x) \bar{f}(x) dx$$

$$\therefore \int_{-\infty}^{\infty} |\hat{f}(w)|^2 dw = \int_{-\infty}^{\infty} |f(x)|^2 dx$$

Parseval's identity for Cosine Transform

$$i. \frac{2}{\pi} \int_0^{\infty} \hat{f}_c(w) \hat{g}_c(w) dw = \int_0^{\infty} f(x) g(x) dx$$

$$ii. \frac{2}{\pi} \int_0^{\infty} |\hat{f}_c(w)|^2 dw = \int_0^{\infty} |f(x)|^2 dx$$

Parseval's identity for sine transform

$$i. \frac{2}{\pi} \int_0^{\infty} \hat{f}_s(w) \hat{g}_s(w) dw = \int_0^{\infty} f(x) g(x) dx$$

$$\text{ii. } \frac{2}{\pi} \int_0^{\infty} |\hat{f}_s(w)|^2 dw = \int_0^{\infty} |f(x)|^2 dx$$

10. Using Parseval's identity show that $\int_0^{\infty} \frac{x^2 dx}{(x^2 + 1)^2} = \frac{\pi}{4}$

Let $f(x) = \frac{x}{x^2 + 1}$ so that

$$\mathfrak{F}_s(f) = \frac{\pi}{2} e^{-w} \quad (\text{proved already from example 4 Page 132})$$

By Parseval's identity for Sine transformation

$$\frac{2}{\pi} \int_0^{\infty} |\mathfrak{F}_s(f)|^2 dw = \int_0^{\infty} |f(x)|^2 dx$$

$$\text{or, } \int_0^{\infty} \left| \frac{x}{x^2 + 1} \right|^2 dx = \frac{2}{\pi} \int_0^{\infty} \left| \frac{\pi}{2} e^{-w} \right|^2 dw$$

$$= \frac{2}{\pi} \frac{\pi^2}{4} \int_0^{\infty} |e^{-2w}| dw$$

$$= \frac{\pi}{2} \left| \frac{e^{-2w}}{-2} \right|_0^{\infty}$$

$$= \frac{\pi}{2} \left| 0 + \frac{1}{2} \right|$$

$$= \frac{\pi}{4}$$

11. Find the Fourier transform of $f(x) = \begin{cases} 1 - |x| & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$

and hence find the value of $\int_0^{\infty} \frac{\sin^4 t}{t^4} dt$.

$$\mathfrak{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [1 - |x|] e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (1 - |x|) (\cos wx + i \sin wx) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1 - |x|) \cos wx dx + \frac{i}{\sqrt{2\pi}} \int_{-1}^1 (1 - |x|) \sin wx dx$$

even function odd function

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 (1 - x) \cos wx dx + 0$$

$$\begin{aligned}
 &= \sqrt{\frac{2}{\pi}} \left[\left\{ (1-x) \frac{\sin wx}{w} \right\}_0^1 + \int_0^1 \frac{\sin wx}{w} dx \right] \\
 &= \sqrt{\frac{2}{\pi}} \left[0 + \left\{ \frac{-\cos wx}{w^2} \right\}_0^1 \right] \\
 &= \sqrt{\frac{2}{\pi}} \left(\frac{1 - \cos w}{w^2} \right)
 \end{aligned}$$

Using Parseval's identity, we get

$$\begin{aligned}
 \int_{-\infty}^{\infty} |\hat{f}(w)|^2 dw &= \int_{-\infty}^{\infty} |f(x)|^2 dx \\
 \text{or, } \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos w)^2}{w^4} dw &= \int_{-1}^1 (1 - |x|)^2 dx \\
 \text{or, } \frac{4}{\pi} \int_0^{\infty} \frac{\left(1 - 1 + 2\sin^2 \frac{w}{2}\right)^2}{w^4} dw &= \int_{-1}^1 (1 + x^2 - 2x) dx
 \end{aligned}$$

LHS is even function, $2x$ on the RHS is odd function

$$\text{or, } \frac{16}{\pi} \int_0^{\infty} \frac{\sin^4 \frac{w}{2}}{w^4} dw = 2 \int_0^1 (1 + x^2) dx = 2 \left(x + \frac{x^3}{3} \right)_0^1$$

$$\text{or, } \frac{16}{\pi} \int_0^{\infty} \frac{\sin^4 \frac{w}{2}}{w^4} dw = \frac{8}{3}$$

Let $\frac{w}{2} = t$, then we get

$$\begin{aligned}
 \frac{16}{\pi} \int_0^{\infty} \frac{\sin^4 x}{16t^4} 2dt &= \frac{8}{3} \\
 \Rightarrow \int_0^{\infty} \frac{\sin^4 t}{t^4} dt &= \frac{4\pi}{3}
 \end{aligned}$$

Exercise 6.3

1. Find the Fourier transform of

- $f(x) = \begin{cases} e^{-kx} & \text{if } x > 0 \quad k > 0 \\ 0 & \text{if } x < 0 \end{cases}$
- $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
- $f(x) = \begin{cases} xe^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$

$$\begin{aligned} \text{d. } f(x) &= \begin{cases} e^{-2x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \\ \text{e. } f(x) &= \begin{cases} -1 & \text{if } -a < x < 0 \\ 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases} \\ \text{f. } f(x) &= \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

2. Find the Fourier sine transform of e^{-x} ($x > 0$) and hence show

$$\text{that } \int_0^{\infty} \frac{x \sin mx}{x^2 + 1} dx = \frac{\pi}{2} e^{-m}$$

3. Find the Fourier cosine transform of e^{-x} and show

$$\int_0^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}.$$

4. Solve:

$$\int_0^{\infty} f(t) \sin wt dt = \begin{cases} 1 & \text{for } 0 \leq t < 1 \\ 2 & \text{for } 1 \leq t < 2 \\ 0 & \text{for } t \geq 2 \end{cases}$$

Answers

$$\begin{aligned} 1. \text{ a. } & \frac{k - iw}{\sqrt{2\pi} (k^2 + w^2)} \\ \text{ b. } & \frac{1}{\sqrt{2\pi}} \left[-\frac{2}{w^2} + \frac{e^{iw}}{iw} - \frac{e^{-iw}}{iw} + \frac{e^{iw}}{w^2} + \frac{e^{-iw}}{w^2} \right] \\ \text{ c. } & \frac{1}{\sqrt{2\pi} (1 + iw)^2} \\ \text{ d. } & \frac{1}{\sqrt{2\pi} (2 + iw)} \\ \text{ e. } & i\sqrt{\frac{2}{\pi}} \frac{(\cos w - 1)}{w} \\ \text{ f. } & \frac{1}{w^2 \sqrt{2\pi}} [e^{-iaw} (1 + iaw) - 1] \end{aligned}$$

$$2. \sqrt{\frac{2}{\pi}} \left(\frac{w}{1 + w^2} \right)$$

$$3. \sqrt{\frac{2}{\pi}} \left(\frac{1}{1 + w^2} \right)$$

$$4. f(x) = \frac{2}{\pi x} (1 + \cos x - 2\cos 2x)$$